K-EMBEDDINGS OF A NORMAL ALGEBRAIC EXTENSION DIFFER BY AUTOMORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The reason Galois groups act simply transitively on embeddings of normal extensions, and the backbone of the extension theorem for automorphisms.

Editorial note. The label lemma-normal-embeddings-differ-by-aut names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Chapter lemma-lift-maps (delivered card L48358, item 2): if L over K is normal algebraic, every K-algebra map from an intermediate extension M into L extends to an automorphism of L
- Chapter lemma-characterize-normal (source lemma): normality is equivalent to all K-embeddings of the extension into a common overfield having one and the same image

Source reference. Original reference 0BME: sources/stacks-fields/pinned-original-source.tex, source lines 2163–2163. Prerequisite pointer invoked at 2252; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let L/K be an algebraic normal extension of fields. Let E/K be an extension of fields. Then either there is no K-embedding from L to E or there is one $\tau : L \to E$ and every other one is of the form $\tau \circ \sigma$ where $\sigma \in Aut(L/K)$.*

Proof. Given τ replace L by $\tau(L) \subset E$ and apply Lemma 0BME. □

Date: stacks-project:fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.